

FMC Data Acquisition and DAS/TFM Post-Processing Workflow

Angled Point-Scatterer NDT Simulation Using the Verasonics Platform

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Abstract

This document describes the acquisition, data saving, and post-processing workflow used for full matrix capture (FMC) ultrasonic imaging of simulated angled point scatterers. The acquisition sequence is implemented on the Verasonics platform using a 32-element linear array operated at a nominal frequency of 5 MHz. In each FMC frame, one array element is activated at a time for transmission while all receive elements are active, thereby generating the complete set of transmit–receive channel combinations required for total focusing method (TFM) or delay-and-sum (DAS) reconstruction. Thirty frames are acquired. The Verasonics reconstruction is retained as a reference complex IQ image, while the raw RF receive data are independently reconstructed using a custom DAS/TFM pipeline in MATLAB or Google Colab. The post-processing procedure includes receive-buffer extraction, FMC cube formation, analytic-signal generation, automatic transmit/receive ordering verification, sample-timing calibration, full-resolution DAS/TFM beamforming, envelope detection, logarithmic compression, and comparison against the Verasonics IQ reconstruction.

1 Purpose and Processing Overview

The workflow has two major parts:

1. **Verasonics acquisition and reference reconstruction:** simulate/acquire FMC RF data, reconstruct a Verasonics IQ image, and save the raw acquisition structures together with the reconstructed in-phase and quadrature data.
2. **Independent post-processing:** reconstruct the same FMC data using a custom DAS/TFM implementation and compare it with the Verasonics IQ image.

The complete processing chain is summarized in Fig. 1.

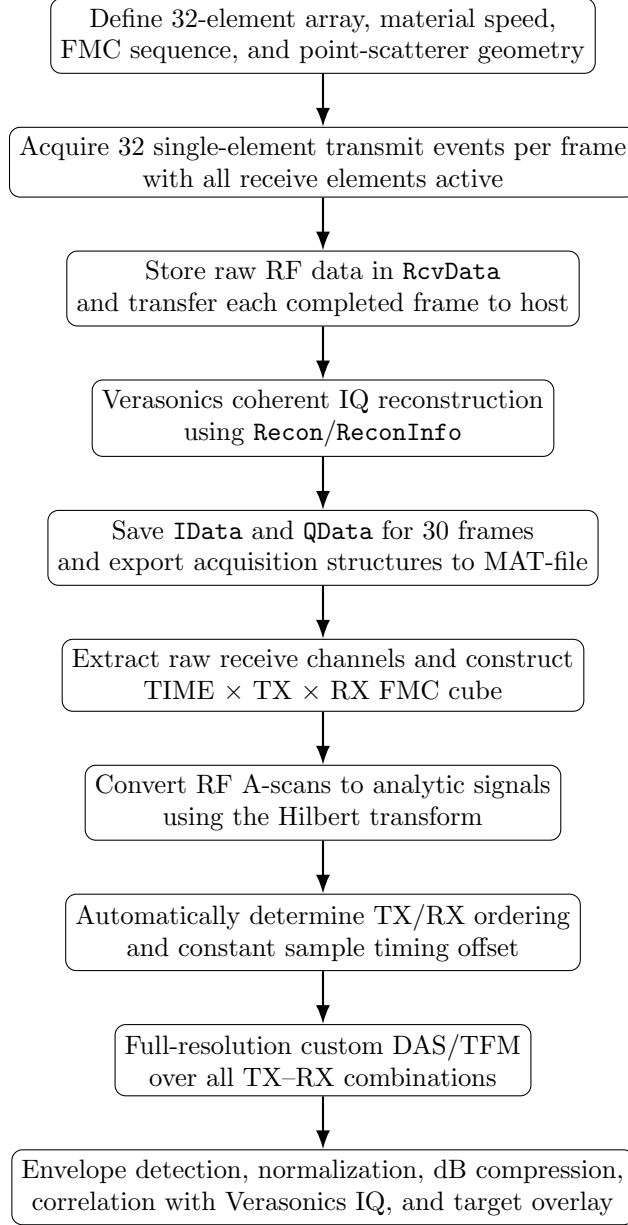


Figure 1: Acquisition and post-processing workflow used for the angled point-scatterer FMC dataset.

2 Acquisition and Simulation Configuration

2.1 Ultrasonic Array and Propagation Parameters

The acquisition script defines a custom 32-element linear array. The principal parameters are listed in Table 1.

Table 1: Acquisition and transducer parameters used in the Verasonics setup.

Parameter	Value / implementation
Number of transmit channels	32
Number of receive channels	32
Number of physical elements	32
Nominal center frequency	5 MHz
Configured fractional bandwidth	0.67
Element width	0.52 mm
Element pitch/spacing	0.60 mm
Array geometry	Linear
Speed of sound	6320 m/s
Transmit voltage setting	10 V
Acquisition start depth	0 wavelengths
Acquisition end depth	150 wavelengths
Receive sample mode	NS200BW
TGC control points	Eight values, all set to 1000
Frames acquired	30
Time between successive FMC transmit events	100 μ s

The acoustic wavelength is

$$\lambda = \frac{c}{f_0}, \quad (1)$$

where $c = 6320$ m/s and $f_0 = 5$ MHz. Therefore,

$$\lambda = 1.264 \text{ mm}. \quad (2)$$

The transducer element positions are generated from the specified 0.60 mm spacing and are centered about the array origin.

2.2 Point-Scatterer Geometry

The executable scatterer definition uses three angles,

$$\theta \in \{40^\circ, 55^\circ, 70^\circ\}, \quad (3)$$

and three radial distances,

$$r \in \{45, 65, 85\} \text{ mm.} \quad (4)$$

Thus, the present configuration generates a 3×3 set of **nine point scatterers**. A comment in the setup file still refers to “16 point scatterers,” but the active arrays in the code contain only three angles and three radii; therefore, the actual executed geometry contains nine targets.

The angles are measured from the positive depth direction ($+Z$), i.e., from the array normal. For each target,

$$x = r \sin \theta, \quad (5)$$

$$z = r \cos \theta. \quad (6)$$

The corresponding target coordinates are listed in Table 2.

Table 2: Point-scatterer locations generated by the active setup code.

Target	θ (deg)	r (mm)	x (mm)	z (mm)
1	40	45	28.93	34.47
2	40	65	41.78	49.79
3	40	85	54.64	65.11
4	55	45	36.86	25.81
5	55	65	53.24	37.28
6	55	85	69.63	48.75
7	70	45	42.29	15.39
8	70	65	61.08	22.23
9	70	85	79.87	29.07

The coordinates are converted from millimetres to wavelengths before being assigned to `Media.MP`. Each point is assigned an amplitude of 1.0.

2.3 FMC Acquisition Sequence

Full matrix capture is implemented using 32 transmit events per frame. For transmit event n , the origin is assigned to the position of element n , and the transmit apodization vector contains a value of one only at that element. Hence, only one element is active for each transmission.

All 32 receive elements are enabled for every transmit event. Therefore, one FMC frame nominally contains

$$N_{\text{paths}} = N_{\text{TX}} N_{\text{RX}} = 32 \times 32 = 1024 \quad (7)$$

transmit–receive propagation paths.

The event loop performs all 32 acquisitions for a frame and then transfers the completed frame to the host. This sequence is repeated for 30 frames.

2.4 Receive Buffer and Reconstruction Grid

The receive buffer is configured as 16-bit integer data. The number of rows per frame is defined as

$$2560 \times 32 = 81920 \quad (8)$$

rows per frame in the Verasonics receive buffer.

The image reconstruction grid is defined by `PData`. Its pixel spacing is specified in wavelength units as

$$\text{PDelta} = [\text{Trans.spacing}, 0, 0.5]. \quad (9)$$

Importantly, the independent DAS/TFM code does not create an unrelated reconstruction grid. Instead, it reads `PData.Origin`, `PData.Size`, and `PData.PDelta` directly from the saved dataset so that the custom reconstruction is evaluated on the same spatial grid as the Verasonics reference image.

3 Verasonics IQ Reconstruction and Data Saving

3.1 Coherent IQ Accumulation

The Verasonics reconstruction uses one `ReconInfo` structure for each synthetic-aperture acquisition. The first transmit event uses `replaceIQ`; intermediate acquisitions use coherent IQ accumulation; and the final transmit event uses `accumIQ_replaceIntensity`. Consequently, complex in-phase and quadrature contributions from the synthetic-aperture acquisitions are accumulated before image intensity formation.

3.2 Saving I and Q Data

An external processing function, `IQ_save`, receives the reconstructed in-phase and quadrature data from the Verasonics intermediate buffer. For each processed frame,

$$\text{IDatasaved}\{k\} \leftarrow \text{IData}, \quad (10)$$

$$\text{QDatasaved}\{k\} \leftarrow \text{QData}. \quad (11)$$

The counter is reset after 30 stored frames, matching the configured number of acquired frames.

The saved complex IQ image is subsequently formed as

$$IQ(x, z) = I(x, z) + jQ(x, z). \quad (12)$$

Its magnitude image is

$$A_{IQ}(x, z) = |IQ(x, z)|, \quad (13)$$

and the logarithmic B-mode representation is obtained from the magnitude in decibels.

3.3 Exported Dataset

The extraction/saving script stores the following variables in a MATLAB `-v7.3` file:

`Resource, RcvData, Trans, PData, Receive, P, Media, TW, TX,`
`IDatasaved, QDatasaved.`

This preserves both the raw receive data and the Verasonics reference reconstruction, together with the geometry and timing metadata needed for independent reconstruction.

4 Independent Post-Processing and Custom DAS/TFM Reconstruction

The custom MATLAB and Google Colab implementations follow the same reconstruction logic. The major processing stages are described below.

4.1 Load the Dataset and Read Acquisition Metadata

For the selected frame, the following quantities are read from the saved structures:

- number of transmit and receive elements;
- speed of sound, c ;
- center frequency, f_0 ;
- receive sampling frequency, f_s , from `Receive(1).decimSampleRate`;
- transmit and receive element positions;
- number of RF samples per acquisition;
- exact Verasonics `PData` grid.

The number of samples in one receive acquisition is determined using

$$N_s = \text{endSample} - \text{startSample} + 1. \quad (14)$$

4.2 Identify the Active Receive-Buffer Columns

In the saved dataset, the raw receive matrix may contain more buffer columns than the 32 physical receive elements. The reconstruction code therefore calculates the energy of every receive-buffer column,

$$E_k = \sum_n |r_k[n]|^2, \quad (15)$$

sorts the columns by energy, and retains the 32 columns with the highest energy. This step identifies the receive-buffer columns carrying the active array data before FMC cube formation.

4.3 Determine the Receive-Buffer Row Organization

Two possible organizations of the raw receive buffer are tested:

- (a) **Compact organization:** the 32 acquisitions are assumed to be stored consecutively with no unused rows between them.
- (b) **Fixed-slot organization:** the full frame is divided into 32 equal transmit slots, and the first N_s samples of each slot are used.

A candidate FMC cube is constructed for each organization, and the total energy is computed. The higher-energy organization is selected automatically.

The resulting FMC array is arranged as

$$\mathbf{R} \in \mathbb{R}^{N_s \times N_{TX} \times N_{RX}}, \quad (16)$$

with the dimensions interpreted as

$$\mathbf{TIME} \times \mathbf{TX} \times \mathbf{RX}.$$

4.4 Recover the Physical TX and RX Coordinates

Receive positions are read from `Trans.ElementPos`, whereas transmit coordinates are obtained from the actual `TX(tx).Origin` values. The transmit origins are stored in wavelength units and are converted to physical distance using the acoustic wavelength.

Using the stored coordinates is important because the independent beamformer must reproduce the same acquisition geometry used by Verasonics.

4.5 Use the Exact Verasonics Reconstruction Grid

For the lateral coordinate,

$$x_i = [x_0 + i \Delta x] \lambda, \quad (17)$$

where $x_0 = \text{PData.Origin}(1)$ and $\Delta x = \text{PData.PDelta}(1)$.

Similarly, for depth,

$$z_j = [z_0 + j \Delta z] \lambda. \quad (18)$$

A 2-D mesh (X, Z) is then constructed. This ensures that the custom DAS/TFM result and the Verasonics IQ image have directly comparable pixel coordinates.

4.6 Construct the Verasonics Reference IQ Image

The saved reference is formed as

$$IQ_{\text{ref}} = I_{\text{saved}} + jQ_{\text{saved}}, \quad (19)$$

followed by envelope calculation and normalization,

$$A_{\text{ref}} = \frac{|IQ_{\text{ref}}|}{\max |IQ_{\text{ref}}|}. \quad (20)$$

This image is used only as a reference for checking raw-data ordering and timing consistency and for final comparison.

4.7 Convert Raw RF A-Scans to Analytic Signals

The raw real-valued RF traces are converted to analytic signals using the Hilbert transform along the time/sample dimension:

$$r_a(t) = r(t) + j\mathcal{H}\{r(t)\}, \quad (21)$$

where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform.

The resulting analytic FMC dataset is complex-valued and retains the phase information required for coherent DAS/TFM summation.

4.8 Automatic TX/RX Ordering Verification

Because the storage direction of the raw FMC data may not necessarily match the physical element-coordinate ordering, four combinations are tested:

1. TX normal / RX normal,
2. TX reversed / RX normal,
3. TX normal / RX reversed,
4. TX reversed / RX reversed.

For computational efficiency, this test is performed on a coarse reconstruction grid and with a subset of array elements. For each ordering, a trial DAS image is reconstructed. The trial image and the Verasonics reference image are converted to dB scale, clipped at -40 dB, and their correlation is calculated inside the selected target-containing region.

The TX/RX ordering with the highest correlation is retained for the final reconstruction.

4.9 Automatic Constant Sample-Timing Calibration

After the best element ordering is selected, a constant sample offset is searched over

$$\Delta n \in [-25, 25] \quad (22)$$

samples.

For every candidate offset, a coarse DAS image is reconstructed using

$$n_{\text{sample}} = \tau f_s + 1 + \Delta n, \quad (23)$$

and correlated with the reference Verasonics IQ image after -40 dB clipping. The offset yielding the maximum correlation is retained as the calibrated timing correction.

The equivalent two-way depth shift corresponding to a sample offset is

$$\Delta z_{2\text{way}} = \frac{\Delta n c}{2f_s}. \quad (24)$$

This calibration compensates for a constant difference between the simple geometric propagation-time model and the timing convention represented in the stored Verasonics data.

5 Full-Resolution DAS/TFM Beamforming

5.1 Transmit and Receive Propagation Distances

For every image pixel (x, z) and transmit element located at x_{TX} , the transmit distance is

$$d_{\text{TX}} = \sqrt{(x - x_{\text{TX}})^2 + z^2}. \quad (25)$$

For a receive element at x_{RX} ,

$$d_{\text{RX}} = \sqrt{(x - x_{\text{RX}})^2 + z^2}. \quad (26)$$

The total bistatic propagation distance is

$$d_{\text{tot}} = d_{\text{TX}} + d_{\text{RX}}, \quad (27)$$

and the corresponding time of flight is

$$\tau = \frac{d_{\text{tot}}}{c}. \quad (28)$$

5.2 RF Sampling and Interpolation

The continuous-valued sample position is evaluated as

$$n(x, z) = \tau(x, z)f_s + 1 + \Delta n_{\text{best}}, \quad (29)$$

where Δn_{best} is the sample offset obtained during calibration.

Because $n(x, z)$ is generally non-integer, the analytic RF signal is sampled using linear interpolation. Samples falling outside the recorded RF range contribute zero.

5.3 Coherent Summation

For a 32-element FMC acquisition, the full DAS/TFM image is obtained by coherently adding the interpolated complex signal from all 1024 transmit–receive paths:

$$I_{\text{DAS}}(x, z) = \sum_{i=1}^{32} \sum_{j=1}^{32} r_{a,ij} \left(\frac{d_{\text{TX},i}(x, z) + d_{\text{RX},j}(x, z)}{c} \right). \quad (30)$$

The coherent sum remains complex-valued. Therefore, the custom beamformed result is analogous to an IQ-domain reconstruction rather than a simple sum of RF magnitudes.

6 Image Formation and Display

6.1 Envelope Detection and Normalization

The beamformed envelope is

$$A_{\text{DAS}}(x, z) = |I_{\text{DAS}}(x, z)|. \quad (31)$$

It is normalized by its global maximum:

$$\tilde{A}_{\text{DAS}} = \frac{A_{\text{DAS}}}{\max A_{\text{DAS}}}. \quad (32)$$

The same normalization procedure is applied to the Verasonics IQ reference magnitude.

6.2 Logarithmic Compression

The normalized images are converted to decibels using

$$I_{\text{dB}} = 20 \log_{10} (\tilde{A} + \epsilon), \quad (33)$$

where ϵ prevents taking the logarithm of zero.

A 40-dB display dynamic range is used, i.e., the displayed values are limited to approximately

$$-40 \text{ dB} \leq I_{\text{dB}} \leq 0 \text{ dB}. \quad (34)$$

6.3 Comparison with Verasonics IQ

The final custom DAS/TFM image is compared with the Verasonics reference in both linear and logarithmic scales. For quantitative comparison, both dB images are clipped below -40 dB and their pixel-wise correlation coefficient is calculated.

This comparison provides a direct check of whether the independently reconstructed image is spatially and structurally consistent with the Verasonics reconstruction.

6.4 Ground-Truth Scatterer Overlay

When the saved `Media` structure is available, the simulated target coordinates are converted from wavelengths back to millimetres and overlaid on the custom DAS/TFM image. This step verifies the localization of the reconstructed point targets relative to the known simulation ground truth.

7 Processing Summary

The complete workflow can be summarized as follows:

1. Define the 32-element, 5-MHz linear array and the NDT propagation speed.
2. Define the nine point scatterers using three angles and three radial distances.
3. Perform FMC using one active transmit element per event and all receive elements active.
4. Repeat 32 transmit events per frame and acquire 30 frames.
5. Transfer each complete frame to the host.
6. Reconstruct coherent complex IQ data using the Verasonics reconstruction engine.
7. Save the separated I and Q data together with the raw RF receive buffer and acquisition metadata.
8. Select a frame for independent reconstruction.
9. Identify the active raw receive-buffer columns.
10. Determine whether the frame is stored in compact or fixed transmit slots.
11. Reshape the raw data into a $\text{TIME} \times \text{TX} \times \text{RX}$ FMC cube.
12. Read the physical element locations and the exact Verasonics reconstruction grid.
13. Convert each RF A-scan to a complex analytic signal using the Hilbert transform.
14. Test the four possible TX/RX ordering combinations.
15. Select the ordering giving the highest correlation with the Verasonics reference.
16. Search a constant timing offset from -25 to $+25$ samples.
17. Apply the best ordering and timing offset in the final full-resolution reconstruction.
18. Calculate the two-way propagation time for every TX-pixel-RX path.
19. Interpolate the analytic RF data at the calculated sample positions.
20. Coherently sum all 1024 TX-RX contributions to obtain the custom DAS/TFM image.
21. Compute the envelope, normalize, and display the image using a 40-dB dynamic range.
22. Compare the custom reconstruction with the Verasonics IQ reference and overlay the true point-scatterer positions.

8 Files Used in the Workflow

The present description is based on the following project files:

- `SetUp_NDTFMC_RD_32_angledata(2).m` — Verasonics FMC acquisition and simulation setup.
- `IQ_save(1).m` — storage of reconstructed in-phase and quadrature frames.
- `EXTRACTANDDISPLAY_IQ.m` — export of the complete dataset and formation/display of complex IQ data.
- `dasreconstructionfromcustomdas(1).m` — MATLAB implementation of the independent DAS/TFM reconstruction and validation.
- `DASonangledata.ipynb` — Google Colab/Python implementation of the same post-processing strategy.
- `simulated3by3pointscattereratangle.mat` — saved angled point-scatterer FMC dataset.

9 Conclusion

The implemented acquisition is a conventional FMC synthetic-aperture sequence in which each of the 32 elements transmits individually and all receive channels record the corresponding echoes. The Verasonics platform produces a complex IQ reference reconstruction while retaining the raw receive data. The independent post-processing pipeline reconstructs the image directly from the saved FMC RF data using geometric two-way propagation delays and coherent complex summation. Automatic verification of raw-data organization, TX/RX ordering, and constant timing offset is included specifically to improve agreement between the custom DAS/TFM image and the Verasonics reference reconstruction. The resulting workflow provides a reproducible basis for subsequent development and evaluation of conventional, nonlinear, or learning-based beamforming methods.